THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL UNIT TEST SET 1

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Assertion (A): Ignoring air resistance in a model of a falling ball is acceptable for a stone. Reason (R): For a stone, the effect of air resistance is very small compared with its weight. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. Assertion (A): Milk shows the Tyndall effect but salt solution does not. Reason (R): The particles of milk are larger than the particles of a true solution. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. Assertion (A): The law of constant proportions is true for compounds but not for mixtures. Reason (R): In a compound the elements are chemically combined in a fixed ratio. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. Assertion (A): Most alpha particles pass undeflected through a thin gold foil. Reason (R): The nucleus occupies a very small fraction of the atom's volume. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. The mass of the Earth is about 6 x 10^24 kg and that of the Sun about 2 x 10^30 kg. The Sun is about ___ times heavier. 1

(a) 3 x 10^5 **(b)** 3 x 10^4 **(c)** 3 x 10^54 **(d)** 3 x 10^6

6. A solution has 30 g of solute in 120 g of solvent. 90 g of the solvent is evaporated. The new percentage by mass of solute is: 1

(a) 50 per cent **(b)** 25 per cent **(c)** 20 per cent **(d)** 75 per cent

7. A compound of A and B contains 75 per cent A by mass. A second compound of the same elements contains 60 per cent A. The ratio of the masses of A that combine with a fixed mass of B in the two compounds is: 1

(a) 3 : 1 **(b)** 3 : 2 **(c)** 4 : 3 **(d)** 2 : 1

8. An ion has 18 electrons and a nucleus with 16 protons. The ion is: 1

(a) S^{2-} **(b)** Ar **(c)** Cl^{-} **(d)** S^{2+}

9. A scientist says 'The model fits in 99 of 100 trials, so the theory is proven'. A better statement is: 1

(a) the theory is false

(b) the theory is strongly supported but remains open to tests

(c) the data are wrong

(d) the theory is proven forever

10. A mixture can be separated by sublimation if: 1

(a) only one component sublimes **(b)** both sublime at equal rates

(c) neither component melts **(d)** both melt

11. A student heats 10 g of CaCO3. Only 5.6 g of solid CaO is left. The mass of CO2 evolved is: 1

(a) 4.4 g **(b)** 15.6 g **(c)** 10 g **(d)** 5.6 g

12. An unknown element has isotopes 24 (79 per cent), 25 (10 per cent) and 26 (11 per cent). The average atomic mass is: 1

(a) 25 u **(b)** 24.9 u **(c)** 24.3 u **(d)** 24 u

13. Assertion (A): A scientific theory can never be changed. Reason (R): A theory is supported by a large amount of evidence. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

14. Assertion (A): Chromatography can separate the colours of a black ink. Reason (R): Different dyes have different solubilities in the solvent. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

15. Assertion (A): The mass of a gas-releasing reaction in an open vessel appears to decrease. Reason (R): Mass is destroyed during reactions in open vessels. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. The sides of a rectangle are measured as 12.0 cm and 8.0 cm with an uncertainty of 0.1 cm each. Find the area and the percentage uncertainty. 2

17. A solution has density 1.2 g/mL and contains 24 g of solute in 200 mL. Find the mass by mass percentage. 2

18. Find the mass of water formed when 4 g of hydrogen reacts with 40 g of oxygen. Which reactant is left over and how much? 2

19. Find the percentage of the heavier isotope for an element of average mass 63.5 u with isotopes 63 u and 65 u. 2

20. Why does a log-log or straight-line graph help in checking a mathematical relationship? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A student measures a pencil three times as 14.2 cm, 14.3 cm and 14.1 cm. Find the mean length and the greatest deviation from it. 3

22. The solubility of a salt is 25 g per 100 g water at 20 degrees Celsius and 60 g per 100 g water at 60 degrees Celsius. Find the mass of crystals obtained when 320 g of saturated solution at 60 degrees Celsius is cooled to 20 degrees Celsius. 3

23. A compound contains 1.6 g of oxygen for 1 g of sulphur (S = 32, O = 16). Find the atoms ratio and the formula. 3

24. A metal M forms an oxide M_2O_3. Its atom has 13 protons. State its configuration, ion and the formula unit mass if its mass number is 27. 3

OR

Explain why a scientist should report measurement uncertainties along with results with an example. 3

25. A student prepared two solutions: 5 g sugar in 50 g water (A) and 8 g sugar in 100 g water (B). Compare the concentrations. 3

OR

Explain why a compound with ions carrying greater charge such as MgO has a higher melting point than NaCl. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Design an investigation to find the relation between the length and the time period of a simple pendulum: apparatus, variables, procedure, table, graph and conclusion. 5

OR

A student analyses a mixture of NaCl (solubility 36 g per 100 g water, almost constant) and KNO3 (solubility 110 g per 100 g at 80 degrees Celsius and 32 g at 20 degrees Celsius). A mixture of 36 g NaCl and 110 g KNO3 is dissolved in 100 g water at 80 degrees Celsius and cooled to 20 degrees Celsius. Find the mass of each solid crystallising and comment on purity. 5

27. A student burns 2.4 g of magnesium in a crucible and finds the oxide mass to be 4.0 g. She repeats with 6.0 g and gets 10.0 g. Show that the results support the law of constant proportions, find the ratio Mg : O and verify the formula MgO (Mg = 24, O = 16). 5

OR

The atomic masses of two isotopes of an element are 69 u and 71 u and its average atomic mass is 69.8 u. Find the percentage abundance of each isotope and the number of neutrons in each if Z = 31. Name the element. 5

END OF QUESTION PAPER Total: 50 Marks